THE UK UNIVERSITY INTEGRATION BEE 2025/2026

$$\int$$

ROUND TWO

GROUP ROUND

Sponsored by



Jane Street

Problems

1. $\displaystyle\int_0^1 \frac{\ln\left(\sin^{-1}(x)\right)}{\sqrt{(1-x^2)\sin^{-1}(x)}}\,\mathrm{d}x$
2. $\displaystyle\int_0^{\pi} \ln\left(\cos(x)+\sqrt{1+\cos^2(x)}\right)\mathrm{d}x$
3. $\displaystyle\int_0^1 \frac{1}{\sqrt[3]{\sqrt[3]{x}-x}}\,\mathrm{d}x$
4. $\displaystyle\int_0^1 \frac{2x-1}{\sqrt{2x-x^2}-\sqrt{1-x^2}}\,\mathrm{d}x$
5. $\displaystyle\int_1^{\infty} \{x\}\left\lfloor\frac{2}{\{x\}}\right\rfloor^{-\lfloor x\rfloor}\mathrm{d}x$
6. $\displaystyle\int_0^{\frac{\pi}{2}} (\sin(x)+\cos(x))\tan^{-1}(\sin(x))\,\mathrm{d}x$
7. $\displaystyle\int_{-\infty}^{\infty} \frac{\cos^2(x)}{\cos^2(x)+(x\cos(x)+\sin(x)+1)^2}\,\mathrm{d}x$
8. $\displaystyle\int_{-\infty}^{\infty} \frac{1}{(e+e^x)(x^4-4x^3+6x^2-4x+2)}\,\mathrm{d}x$
9. $\displaystyle\int_0^{\infty} \frac{\sin^6(x)}{x^2}+\frac{\sin^5(x)}{x}+\frac{\sin^4(x)}{x^2}+\frac{\sin^3(x)}{x^3}\,\mathrm{d}x$
10. $\displaystyle\int_0^{\infty} \frac{x(2x+3)\sqrt[3]{x+1}}{x^6+x^2+2x+1}\,\mathrm{d}x$
11. $\displaystyle\left\lfloor\int_0^1 \pi(e^{\pi x}-1)+(1+x)(1-x+x^2-x^3+\cdots+x^{998})\,\mathrm{d}x\right\rfloor$
12. $\displaystyle\sum_{n=1}^{\infty}\int_0^1 (-1)^n \ln\left(\frac{1-x^{\frac{1}{n}}}{1-x^{\frac{1}{n+1}}}\right)\mathrm{d}x$
13. $\displaystyle\left\lceil\int_1^{\infty} \frac{1}{\tan^{-1}(x)}-\frac{1+x^2}{1+\tan^{-1}(x)+x^2\tan^{-1}(x)}\,\mathrm{d}x\right\rceil$
14. $\displaystyle\int \frac{(x+1)^2+(3x+1)\sqrt{x+\ln(x)}}{x\sqrt{x+\ln(x)}\left(x+\sqrt{x+\ln(x)}\right)}\,\mathrm{d}x$
15. $\displaystyle\int_0^1 \sum_{n=1}^{\infty} x^n \lim_{t\to 0} f_n(t)\,\mathrm{d}x$, where $\displaystyle f_n(t)=\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}}\left[\frac{t^n}{n!\sin^n(t)}\right]$.